Semester Test 1 Memo

Please read the following carefully.

1. Answer all the questions in the space provided on the question paper.
2. No pencil work will be marked. Do not use "tippex".
3. If you need more space, please use the pages back of the previous or current page, clearly labelling your answer on these pages. Also clearly instruct the marker to find the rest of your answer on a specific page.
4. Where applicable you need to draw the necessary diagrams (e.g. vector diagrams with axes, free-body diagrams).
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.**
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. The use of the cheat sheet is only allowed subject to the rules stated on it. You may only write on one side of your cheat sheet.
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

Constants and formulas
$G = 6.674 \times 10^{-11}$ N·m^2/kg^2; $e = 1.602 \times 10^{-19}$ C; $k = 1.381 \times 10^{-23}$ J/K; $m_e = 9.109 \times 10^{-31}$ kg; $m_p = 1.673 \times 10^{-27}$ kg; $R = 8.314$ J/(mol·K); $N_A = 6.02 \times 10^{23}$ /mol; 1 atm = 101.3 kPa; 1 u = $1.6605677 \times 10^{-27}$ kg; $c = 3 \times 10^8$ m/s; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; 1 litre = 1000 cm^3; $M_{\text{Earth}} = 5.97 \times 10^{24}$ kg, $\rho(\text{water}) = 1$ g/cm^3; $r_{\text{Earth}} = 6.37 \times 10^6$ m

1 Question 1

[14]

(a) An old English children's rhyme states, "Little Miss Muffet sat on a tuffet, eating her curds and whey. Along came a spider and sat down beside her, and frightened Miss Muffet away." The story goes that the spider sat down next to Miss Muffet, not because of her curds and whey, but because of the 11 tuffets of dried flies. The volume measure of a tuffet is given by 1 tuffet = 2 pecks = 0.5 bushel, where an Imperial bushel = 36.3687 liters (L). What was Miss Muffet's stash in liters? (3)

3

$$11 \text{ tuffets} \times \frac{0.5 \text{ bushel}}{\text{tuffet}} \times \frac{36.3687 \text{ L}}{\text{bushel}}$$

$= 200$ L

(b) A drinking glass is designed to hold 250 ml of fluid. The radius, r cm, of the glass is constant and the height is h cm. (11)

i. Write an expression for h as a function of the radius r. (2)

$V = \text{Area} \times \text{height} = \pi r^2 h$

$h = \dfrac{V}{\pi r^2}$

2

ii. Calculate the **interior** surface area, A, of the glass as a function of r only. (4)

Area: $A = \text{bottom area} + \text{sides}$

$A = \pi r^2 + (2\pi r h)$

$= \pi r^2 + 2\pi r \left(V/\pi r^2\right)$

$= \pi r^2 + 2\cancel{\pi r}\left(\dfrac{250}{\cancel{\pi} r^{\cancel{2}}}\right)$

$= \dfrac{500}{r} + \pi r^2$

4

iii. Calculate the value of r for which A is a minimum and then determine the minimum value of A. (5)

A will be a minimum when $\dfrac{dA}{dr} = 0$

$\dfrac{dA}{dr} = \dfrac{d}{dr}\left[\dfrac{500}{r} + \pi r^2\right] = 0$

$0 = -500 r^{-2} + 2\pi r$

$\dfrac{500}{2\pi} = r^3$

$\therefore \; r = 4.3 \text{ cm}$

$A = \dfrac{500}{4.3(\text{cm})} + \pi(4.3 \text{ cm})^2$

$= 175 \text{ cm}^2$

5

11

2 Question 2

[8]

A key falls from a bridge that is 45 m above the water. It falls directly into a model boat, moving with constant velocity, that was 12 m from the point of impact when the key is released.

(a) What is the speed of the boat? (5)

Key: $\Delta y = -45\,m$ $\quad a_y = -9.8\,m/s$ $\quad V_{oy} = 0\,m/s$

boat: $\Delta x = 12\,m$ $\quad V_x =$

Time for key to fall

$\Delta y = V_{oy}t + \frac{1}{2}a_y t^2 = \frac{1}{2}a_y t^2$

$t^2 = \frac{2\Delta y}{a_y} = \frac{2(-45\,m)}{-9.8\,m/s^2}$ Note - signs!

$t = 3.03\,s$

Boat: $\Delta x = V_{ox} t$

$V_{ox} = \frac{\Delta x}{t} = \frac{12\,m}{3.03\,s} = 4.0\,m/s$

(b) Determine the speed of the key as it lands on the boat. (3)

3

$V_y = V_{iy} + a_y t$ OR $V_y^2 = V_{iy}^2 + 2a_y \Delta y$

substitute $a_y = -9.8\,m/s^2$; $\Delta y = -45\,m$

$V_y = -29.7\,m/s \approx -30\,m/s$. or downwards.

3 Question 3

[10]

The position of a particle moving along the x-axis is given by $x = 12t^2 - 2t^3$, where x is in meters and t is in seconds.

(a) Determine the velocity at $t = 3.0$ s. (2)

$V = \frac{dx}{dt} = 24t - 6t^2$

At $t = 3.0\,s$

$V = 24(3.0\,s) - 6(3.0\,s)^2$

$= 18\,m/s$

(b) Determine the maximum position reached by the particle along the x-axis. (5)

5

Max position when $V = 0$, ie $\frac{dx}{dt} = 0$

$V = 24t - 6t^2 = 0$

$\therefore t = 0\,s$ or $4.0\,s$

Position at $t = 4.0\,s$

$x = 12t^2 - 2t^3 = 12(4.0s)^2 - 2(4.0s)^3$

$x = 64\ m$

(c) Determine the particle's acceleration at the time when the maximum position in (b) is reached? (3)

3

$a = \frac{dV}{dt} = 24 - 12t$

At $t = 4.0\ s$

$a = 24 - 12(4.0s)$

$= -24\ m/s^2$

Question 4

[8]

Consider the following two vectors:

$$\vec{a} = (4.0\hat{\imath} - 3.0\hat{\jmath})\text{m}$$

and

$$\vec{b} = (6.0\hat{\imath} + 8.0\hat{\jmath})\text{m}.$$

(a) Determine the magnitude and direction of $\vec{a} - 2\vec{b}$. (4)

$\bar{a} - 2\bar{b} = (4.0\hat{i} - 3.0\hat{j})m - 2(6.0\hat{i} + 8.0\hat{j})m$

$= [(4.0 - 12.0)\hat{i} + (-3.0 - 16.0)\hat{j}]m$

$\bar{c} = (-8.0\hat{i} - 19\hat{j})\,m$

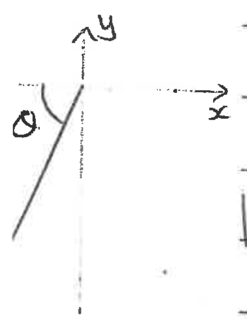
$|\bar{a} - 2\bar{b}| = \sqrt{c_x^2 + c_y^2} = \sqrt{(-8.0m)^2 + (-19.0m)^2}$

$= 20.6\,m \approx 21\,m$

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-19}{-8}\right)$

$= 67°$ (below x-axis)

$\approx 247°$ OR



(b) Determine the angle between $\vec{a}$ and $\vec{b}$. (4)

4

$\bar{a} \cdot \bar{b} = |a||b| \cos\theta$

$|a| = \sqrt{(4.0)^2 + (-3.0)^2}$ m = 5 m

$|b| = \sqrt{(6.0)^2 + (8.0)^2}$ m = 10 m

$\bar{a} \cdot b = a_x b_x + a_y b_y = |a||b| \cos\theta$

$\cos\theta = \frac{a_x b_x + a_y b_y}{|a||b|} = \frac{(4.0m)(6.0m) + (-3.0m)(8.0}{(5m)(10m)}$

$\theta = 90°$

Question 5

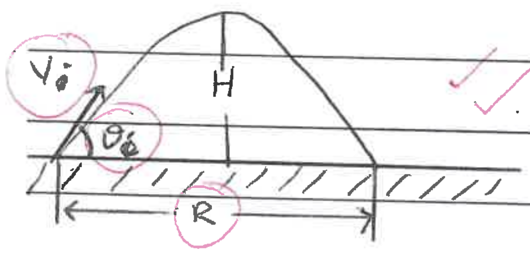
[9]

9

Using the equations of motion $x = x_i + v_{xi}t + \frac{1}{2}a_x t^2$ and $y = y_i + v_{yi}t + \frac{1}{2}a_y t^2$, show that the range of a projectile's motion is given by

$$R = \frac{v_i^2}{g}\sin 2\theta_i,$$

where θ_i is the launch angle and v_i is the launch velocity. Clearly state all assumptions, rules, laws and mathematical relations you may use. **Draw a neat sketch and label the important variables on the sketch**. The following identity may be useful: $2\sin\theta\cos\theta = \sin 2\theta$.



V_i : initial Velocity

θ_i : launch angle

R : Range

$V_{ix} = V_i \cos\theta_i$ $\quad V_{iy} = V_i \sin\theta_i$

$a_x = 0\ m/s^2$ $\quad a_y = -9.8 m/s^2 = -g$

$\Delta x = R = V_i \cos\theta_i (t)$ $\quad \Delta y = 0$

$t = \frac{R}{V_i \cos\theta_i}$ — ①

$\Delta y = V_{iy} t + \frac{1}{2} a_y t^2$ — ②

① in ② $0 = V_i \sin\theta_i \left(\frac{R}{V_i \cos\theta_i}\right) - \frac{1}{2} g \left(\frac{R}{V_i \cos\theta_i}\right)^2$

Rearranging: $\frac{2 v_i^2 \sin\theta_i \cos\theta_i}{g} = R$

But $2\sin\theta\cos\theta = \sin 2\theta$

Then $R = \frac{V_i^2 \sin 2\theta_i}{g}$

Continue on the next page

Question 6

[11]

A ball is shot from the ground into the air. At a height of 9.1 m, its velocity is $\vec{v} = (7.6\hat{\imath} + 6.1\hat{\jmath})$m/s.

v_i

(a) To what maximum height does the ball rise? (4)

4

In y-direction (vertical) $v_{iy} = 6.1\,m/s$ $v_{fy} = 0\,m/s$

$\therefore v_{fy}^2 = v_{iy}^2 + 2a_y \Delta y$

$\Delta y = \dfrac{v_{fy}^2 - v_{iy}^2}{2a} = \dfrac{0 - (6.1\,m/s)^2}{2(-9.8\,m/s)}$

$= 1.9\ m$

Total height: $y_i + 1.9m = 9.1m + 1.9m$

$= 11\ m$

Can also first find v_{yi} then Δy total.

(b) How far horizontally from the launch point will the ball land if the area is flat? (5)

$R = \dfrac{v_0^2 \sin 2\theta_0}{g}$

$v_f^2 = v_{0y}^2 + 2a\Delta y$

$v_{0y}^2 = v_{y\,max}^2 - 2a\Delta y = 0 - 2(-9.8\,m/s^2)(11\,m)$

$v_{0y} = 14.7\,m/s$ $\quad v_0 = \sqrt{(7.6\,m/s)^2 + (14.7\,m/s)^2} = 16.5\,m/s$

$\tan\theta_0 = \dfrac{v_{0y}}{v_{0x}} = \dfrac{14.7\,m/s}{7.6\,m/s}$

$\theta_0 = 62.6°$

$R = \dfrac{(16.5\,m/s)^2 \sin(2(62.6°))}{9.8\,m/s^2}$

$= 23\ m$

Alternative method: Determine time of flight then $\Delta x = R = v_{0x} t$

(c) What are the magnitude and direction of the ball's velocity as it hits the ground? (2)

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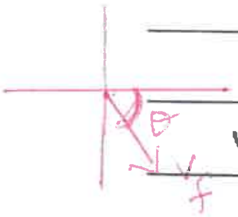
Landing angle = launch angle

$\tan\theta = \frac{-V_{oy}}{V_{ox}} = \frac{-14.7\,m/s}{7.6\,m/s}$



$\theta = -62.6°$ (below $+x$ axis)

$V_0 = V_f = \sqrt{V_{xo}^2 + V_{yo}^2} = 16.5\,m/s$.

Question 7 [9]

A moderate wind accelerates a pebble over a horizontal xy-plane with a constant acceleration $\vec{a} = (5.00\hat{\imath} + 7.00\hat{\jmath})\text{m/s}^2$. At time $t = 0$ s, the velocity is $\vec{v} = (4.00\hat{\imath})\text{m/s}$.

(a) Determine the magnitude and direction of the pebble's velocity when it has been displaced by 12.0 m parallel to the x-axis. (6)

6

$\Delta x = 12.0\,m$ $\quad a_x = 5.00\,m/s^2$ $\quad V_{ox} = 4.00\,m/s$

$a_y = 7.00\,m/s^2$ $\quad V_{oy} = 0\,m/s$

Determine time to reach $\Delta x = 12\,m$

$\Delta x = V_{ox}t + \frac{1}{2}a_x t^2$

Use quadratic soln equn: $t = -3.13\,s$ or $1.53\,s$

$V_x = V_{ox} + a_x t = 4.00\,m/s + 5.00\,m/s^2(1.53\,s)$

$= 11.7\,m/s$

$V_y = V_{oy} + a_y t = 0 + 7.00\,m/s^2(1.53\,s)$

$= 10.7\,m/s$

$V = \sqrt{(11.7\,m/s)^2 + (10.7\,m/s)^2} = 15.8\,m/s$

$\theta = \tan^{-1}\left(\frac{10.7\,m/s}{11.7\,m/s}\right) = 42.4°$ above $+x$ axis

(b) Determine the displacement along the y-direction when the pebble has moved 12.0 m parallel to the x-axis. (3)

3

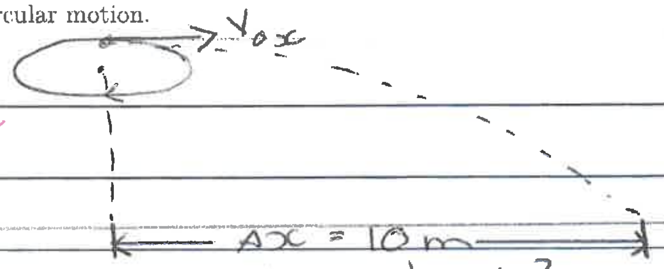
$\Delta y = V_{oy}t + \frac{1}{2}a_y t^2$

$= 0 + \frac{1}{2}(7.00\,m/s^2)(1.53\,s)^2$

$= 8.19\,m$

8 **Question 8** [8]

A boy whirls a stone in a horizontal circle of radius 1.5 m and at a height of 2.0 m above the ground. The string breaks, and the stone flies off **horizontally** and strikes the ground after travelling a horizontal distance of 10 m. Determine the magnitude of the centripetal acceleration of the stone while in circular motion.



$y = 2.0$ m

$\Delta y = -2.0$ m

$y = 0$

$\Delta x = V_{ox} t + \frac{1}{2} a_x t^2 = V_{ox} t$

$V_{ox} = \Delta x / t \quad -①$

$\Delta y = V_{oy} t + \frac{1}{2} a_y t^2 = \frac{1}{2} a_y t^2$

$t = \sqrt{\frac{2\Delta y}{a_y}} = \sqrt{\frac{2(-2.0\text{m})}{-9.8\text{m/s}^2}} = 0.64\text{s} \quad -②$

note - signs

② in ①

$V_{ox} = 10\text{m} / 0.64\text{s}$

$= 15.6$ m/s

$a_c = \frac{V^2}{r} = \frac{(15.6\text{ m/s})^2}{1.5\text{m}} = 162 \text{ m/s}^2$

The end!